

Data Card

Civil Comments — training data for the early-intent-detection toxicity probe

Project	Malicious/Harmful Intent Probe (Komadugu Team, Challenge 1)
Dataset	Civil Comments (google/civil_comments on Hugging Face)
Card version	1.0
Prepared by	Komadugu Team

1. Dataset Summary

Civil Comments is a public dataset of roughly 2 million English-language comments originally posted to the independent Civil Comments platform (2015–2017) before it shut down. Each comment was later rated by multiple crowd annotators for perceived toxicity, producing a continuous **toxicity** score in $[0, 1]$ plus six toxicity sub-attribute scores (severe_toxicity, obscene, threat, insult, identity_attack, sexual_explicit). The dataset is distributed via the Hugging Face datasets library and via the Jigsaw "Unintended Bias in Toxicity Classification" Kaggle competition.

2. Motivation — why this dataset

- Publicly available with documented provenance and a permissive license, avoiding legal/ethical risk from scraping.
- Human-rated (not machine-labeled), which avoids circularity/label leakage from another toxicity classifier.
- A standard, widely-cited benchmark for toxicity and identity-bias research, enabling comparison with prior work.
- Directly relevant to the project's goal of detecting harmful intent from language patterns before a response is generated.

3. Composition & Schema

Field	Type	Description
text	string	Raw comment text (variable length; truncated to 64 tokens in this project).
toxicity	float $[0,1]$	Fraction of raters who judged the comment toxic. Binarized here at ≥ 0.5 .
severe_toxicity, obscene, threat, insult, identity_attack, sexual_explicit	float $[0,1]$	Toxicity sub-attributes. Present in the source data; not used by this project's probe.
identity subgroup annotations	float $[0,1]$	Mentions of identity groups (gender, race, religion, etc.), used in the original bias benchmark. Not used here.
Full dataset size	~1.8M train / ~97K validation / ~97K test comments	

Subset used	First 15,000 shuffled train examples and 5,000 shuffled validation examples (seed = 42); full corpus usable by adjusting a config flag
Label used	Binary: label = 1 if toxicity ≥ 0.5 , else 0
Class balance	Imbalanced — toxic comments are a minority (~8% in the full training set), addressed via class-weighted logistic regression

4. Collection Process

Comments were originally authored by the public on the Civil Comments platform, an experiment in crowd-moderated commenting. After the platform's shutdown, the raw comments were released for research and subsequently annotated by paid crowd workers (via Jigsaw / Conversation AI, a Google-affiliated research group), with multiple independent ratings per comment averaged into the toxicity score. This project performs no additional collection — data is loaded programmatically via `datasets.load_dataset("google/civil_comments")`.

5. Preprocessing Applied in This Project

- Shuffle (seed 42) and subsample to 15,000 train / 5,000 validation examples.
- Binarize the continuous toxicity score at a 0.5 threshold to create the classification target.
- Tokenize with the Gemma-2-2B tokenizer (max length 64 tokens, dynamic padding, truncation).
- Convert each comment to a fixed-size vector via frozen Gemma-2-2B layer-14 hidden states, mean-pooled over non-padding tokens — no text is stored beyond this vector for model training.

6. License & Access

Civil Comments is released by Jigsaw/Conversation AI under a CC0 (public domain) designation as documented on its Hugging Face dataset page. Because of its size and open license, it is **not bundled** in this submission; it is instead downloaded and cached automatically at run time via the Hugging Face datasets library. *Teams redistributing derived data should re-verify current license terms on the dataset's hosting page before external release.*

7. Known Limitations & Risks

- Documented identity bias: comments mentioning certain identity terms (e.g., religion, sexual orientation, nationality) are historically over-flagged as toxic by models trained on this data, even when the comment itself is benign — the motivating problem behind the original Jigsaw Unintended Bias challenge.
- English-only and time-bound to 2015–2017 platform language/slang; may not generalize to other languages or current online discourse.
- Class imbalance can bias a naively-trained classifier toward the majority (non-toxic) class.
- Toxicity ratings reflect subjective crowd judgments and may not represent all cultural norms of what is "harmful."
- Comment text may contain offensive, upsetting, or profane language by design, since the task is to detect it.

8. Maintenance & Stability

The dataset is a static, versioned snapshot hosted on the Hugging Face Hub; the original source platform is defunct, so no new comments are being added. This project does not modify or re-host the dataset — it is re-downloaded from the canonical source on each fresh environment via the reproduction scripts.

9. Intended Use in This Project

Training and evaluation data for a linear probe that classifies frozen Gemma-2-2B embeddings as toxic/safe, as an early, low-cost signal of potentially harmful user intent. **Out of scope:** this dataset and probe are not evaluated for multilingual use, are not a substitute for human moderation review, and should not be the sole safety mechanism in a deployed system without the human-oversight and fairness-auditing safeguards described in the project's Impact Statement.

Sources: Borkan et al. (2019), Nuanced Metrics for Measuring Unintended Bias with Real Data for Text Classification, WWW'19 Companion; huggingface.co/datasets/google/civil_comments.